

System Architecture & Feature Documentation

Comprehensive Technical & Operational Reference Manual for Clients & Administrators

Document Version:	2.0 (Production Release)	Target Stack:	Next.js 16 Full-Stack / MongoDB
Release Date:	September 2026	Auth & Security:	JWT Cookie DAL / Mandatory 1st-Login Gate
Platform Scope:	Admin Portal & Customer Portal	Integrations:	SLASH Starlink API, Google Maps, SMTP
Document Purpose:	Client Feature Walkthrough & System Guide	Confidentiality:	Proprietary & Confidential

EXECUTIVE SUMMARY & SYSTEM SCOPE

The **Hybrid Networks Portal** is an enterprise-grade web application engineered to solve end-to-end satellite and telecommunications lifecycle management. It bridges raw wholesale usage feeds (CDRs) to branded client tax invoices, supports complex multi-account organizational hierarchies under a single customer login, delivers real-time Starlink satellite telemetry via the SLASH API, provides live GPS fleet tracking on Google Maps, and enforces strict security with mandatory first-login password rotation.

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1. APPLICATION OVERVIEW

1.1 What the Application Is

The **Hybrid Networks Portal** is a dedicated web-based software suite built for managing satellite communications, maritime vessel connectivity, telecommunications usage records (CDRs), and multi-account customer billing. It provides unified management for network operators and an intuitive self-service portal for corporate end clients.

1.2 The Problem It Solves

- **Automating Complex Wholesale-to-Retail CDR Rating:** Satellite networks generate high-volume raw usage files with diverse product codes (voice, SMS, standard data, prioritized data). Hybrid Networks ingests, maps, and applies custom markup formulas automatically.
- **Multi-Account Consolidation:** Corporate customers frequently hold multiple sub-accounts or vessels across different billing codes. The platform aggregates these under a single customer credential with instant multi-account visibility.
- **Real-Time Satellite Telemetry:** Operators and clients can monitor live Starlink satellite terminal metrics (latency, signal drops, download/upload speeds, SNR) without logging into external vendor consoles.
- **Zero-Loss CDR Reconciliations:** Unrecognized customer codes or unmapped product codes are flagged immediately, generating downloadable Unallocated Reports and allowing one-click reprocessing once mappings are added.
- **Automated Tax-Compliant Invoicing:** Calculates recurring plan charges alongside rated CDR usage to generate client-ready PDF invoices formatted with itemized breakdowns and tax amounts.

1.3 Target Audience & User Personas

User Persona	Primary Responsibilities	Primary System Interactions
Super Admin <i>(Executive / Management)</i>	Full operational oversight, pricing rules, staff management, and system-wide configurations.	Admin Dashboard, Retail Plans, User Management, Activity Audit Logs, System Settings.
Sub Admin <i>(Billing & Support Staff)</i>	Customer account creation, CDR upload processing, invoice generation, ticket resolution.	Customer Management, CDR Ingestion, Invoice Module, Support Desk, Fleet Map.
Customer <i>(Corporate Client / Fleet Mgr)</i>	Self-service usage tracking across vessels, invoice downloads, support ticket creation.	Customer Portal Overview, Device Telemetry, Live GPS Map, Invoices, My Tickets.

2. APPLICATION MODULES

The codebase is structured into nine functional pillars, completely isolated by role-based access middleware:

Module	Scope & Route Boundaries	Key Capabilities
1. Authentication & Security	/login /first-login-change-password	Role dispatch, bcrypt hashing, brute-force lockout, mandatory password reset gate.
2. Customer Management	/admin/customers /admin/customers/[id]	Multi-account code linking, Starlink Vessel ID binding, profile lifecycle (Active/Suspended).
3. CDR Ingestion & Rating	/admin/cdr-import /admin/retail-plans	PapaParse CSV parser, fuzzy column detection, markup calculator, unallocated CDR alert.
4. Invoicing & Billing	/admin/invoices /portal/invoices	Automated monthly batch billing, React-PDF renderer, status flow (Draft, Sent, Paid, Void).
5. Fleet & Telemetry	/admin/tracking /portal/tracking	Live SLASH API Starlink metrics (latency, ping drops, SNR), Google Maps GPS markers.
6. Terminal Inventory	/admin/terminals /portal/overview	Hardware inventory, MAC/IP assignments, connection health badges, status overrides.
7. Service Plans Catalog	/admin/plans	Recurring bandwidth tier configuration, monthly price, data allowances, contention ratio.
8. Support Desk & Tickets	/admin/support /portal/support	Interactive threaded communication, priority queues, status tracking (Open, In Progress, Closed).
9. Settings & Audit Logs	/admin/settings /admin/activity-logs	Company details, SMTP configuration, staff user management, immutable audit trail.

3. USER ROLES & PERMISSION MATRIX

Access control is enforced at both the Next.js Middleware edge proxy and in Server Actions via Data Access Layer (DAL) assertions:

System Capability / Feature	SUPER_ADMIN	SUB_ADMIN	CUSTOMER
Access Admin Dashboard (/admin/*)	[YES] Full Access	[YES] Full Access	[NO] Forbidden
Access Customer Portal (/portal/*)	Staff Boundary	Staff Boundary	Own Data Only
Create / Edit / Suspend Customers	[YES] Full Access	[YES] Full Access	[NO] Forbidden
Assign Multi-Account Codes	[YES] Full Access	[YES] Full Access	[NO] Forbidden
Upload & Ingest Wholesale CDRs	[YES] Full Access	[YES] Full Access	[NO] Forbidden
Export Unallocated CDR CSV Reports	[YES] Full Access	[YES] Full Access	[NO] Forbidden
Configure Retail Plans & Markups	[YES] Full Access	[YES] Edit	[NO] Forbidden
Generate & Void Invoices	[YES] Full Access	[YES] Full Access	Own Data Only
Download PDF Invoices	All Clients	All Clients	Own Data Only
View Global Fleet GPS Map	All Vessels	All Vessels	Own Data Only
Real-Time Starlink Telemetry	All Vessels	All Vessels	Own Data Only
Manage Support Desk Tickets	[YES] Full Access	[YES] Full Access	Own Data Only
Manage Staff Users & Roles	[YES] Full Access	Read Only	[NO] Forbidden
Configure SMTP & Company Settings	[YES] Full Access	Read Only	[NO] Forbidden
Inspect System Activity Audit Logs	[YES] Full Access	Read Only	[NO] Forbidden

4. FEATURE OVERVIEW

Key operational highlights implemented across the application:

- **Dual Domain Separation:** Distinct portal layouts, navigation trees, and authorization boundaries for Admin and Customer views.
- **Multi-Account Hierarchy:** Customers with multiple corporate accounts or vessels see all data combined under a single login without switching sessions.
- **First-Login Security Gate:** Enforces password change on initial login before any protected pages can be accessed.
- **Fuzzy CDR Header Detection:** PapaParse streaming parser recognizes arbitrary CSV header variations (e.g. 'Customer Code', 'Cust_ID', 'Account #').
- **Dual-Formula Rating Engine:** Supports percentage markups (wholesale cost * (1 + markup%)) and fixed-fee billing models.
- **Unallocated CDR Quarantine:** Safeguards against creating fake accounts; unknown codes trigger clear alerts and downloadable CSV reports.
- **One-Click CDR Reprocessing:** Allows unallocated records to be rated immediately after missing mappings or accounts are registered.
- **React-PDF Invoice Engine:** Generates high-definition PDF invoices dynamically with itemized usage breakdowns and tax details.
- **SLASH API Live Telemetry:** Pulls SNR, latency, ping drops, and upload/download speeds for active Starlink terminals.
- **Interactive Fleet Tracking:** Google Maps visualization displaying real-time vessel coordinates, headings, and connection health.

5. DETAILED FEATURE DOCUMENTATION

5.1 Authentication & Mandatory First-Login Security Gate

Implementation: lib/auth/dal.ts, lib/actions/auth-actions.ts, proxy.ts

Workflow: When an administrator registers a customer or staff user, the system issues a temporary password and flags `mustChangePassword = true`. When the user logs in at `/login`, credentials are validated using `bcryptjs`. If `mustChangePassword` is active, the Data Access Layer (DAL) immediately redirects the session to `/first-login-change-password`. The user cannot access any portal page until they provide the temporary password and a valid new password (minimum 8 characters, at least 1 digit). Upon success, the flag is cleared to false and an encrypted HTTP-only JWT cookie (`hn_session`) is established.

5.2 Customer Management & Multi-Account Linking

Implementation: lib/actions/customer-actions.ts, components/admin/CustomerFormModal.tsx

Workflow: Administrators can create, update, suspend, or delete customer accounts. The modal includes a dedicated Customer Account Code(s) field allowing comma-separated codes (e.g., ZZSP100, ACC-2001, VSAT-409). The backend stores these in `User.customerCodes` while maintaining `User.customerCode` as primary. When CDR files are uploaded or usage records queried, any matching code in the array automatically maps to this customer. Admins can also assign a `starlinkVesselId` to link live satellite telemetry directly to the profile.

5.3 CDR File Ingestion & Automatic Detection

Implementation: lib/cdr/parser.ts, components/admin/cdr/CdrUploadZone.tsx

Workflow: Supports both bulk (multi-customer) and individual wholesale CDR CSV files. Files are parsed on-the-fly using `PapaParse`. The `detectColumns()` algorithm dynamically analyzes header aliases (e.g. 'Customer Code', 'CustID', 'Account', 'Wholesale Cost', 'Wholesale_Amt', 'Product Code', 'Traffic Type'). The system calculates a SHA-256 hash of the file content to prevent duplicate batch uploads.

5.4 CDR Rating Engine & Product Code Matching Matrix

Implementation: lib/billing/pricing-engine.ts, models/CdrChargeRecord.ts, models/RetailPlan.ts

Workflow: For each record, the rating engine matches the raw `customerCode` against all `User.customerCodes`. It then evaluates the row's product identifier against active `RetailPlan` mappings. It supports compound product codes (e.g., Type, CALL - CODE-123, Type, SMS - CODE 245). When a match is made, the retail charge is computed: $\text{wholesaleAmount} * (1 + \text{markupPercentage} / 100)$ or fixed price. A permanent snapshot is stored in `CdrChargeRecord` to ensure historical pricing integrity.

5.5 Unallocated Records Alert & Unmatched CDR Reporting

Implementation: app/api/admin/cdr-import/[batchId]/unallocated-report/route.ts, components/admin/cdr/UnallocatedAlert.tsx

Workflow: If a row contains an unknown Customer Code or an unmapped Product Code, the engine marks it as UNMATCHED with a clear reason. It never creates dummy customer accounts. An **Unallocated Records Alert Card** is rendered prominently. The administrator can click **'Download Unallocated Report (.csv)'** to export an audit CSV containing exact row numbers, missing codes, and wholesale amounts. Once the admin registers the customer or maps the product code, clicking **'Reprocess Unmatched'** immediately rates the quarantined records.

5.6 Billing, Invoicing & PDF Generation

Implementation: lib/actions/invoice-actions.ts, lib/pdf/invoice-template.tsx, app/api/invoices/[id]/pdf/route.ts

Workflow: Invoices aggregate monthly recurring subscription fees and matched CDR usage charges for a billing period. Admins can generate invoices in batch or individually. Invoices transition through a defined lifecycle: DRAFT -> SENT -> PAID / VOID. Both admins and customers can generate and download branded PDF invoices rendered on-the-fly via @react-pdf/renderer, complete with company header, tax breakdown (GST/VAT), itemized line items, payment terms, and remittance instructions.

5.7 Terminal Inventory & Live Starlink Telemetry

Implementation: lib/starlink/client.ts, components/portal/StarlinkTelemetryWidget.tsx

Workflow: For customer profiles linked to a starlinkVesselId, the system queries the SLASH API (<https://slash-api.rudra.sh/api/v1>). It retrieves real-time terminal metrics including round-trip latency (ms), ping drop percentages, download throughput (Mbps), upload throughput (Mbps), signal-to-noise ratio (SNR), obstruction status, and active satellite beam state. Visual status badges display connection health with color-coded telemetry gauges.

5.8 Interactive GPS Fleet Tracking Map

Implementation: components/admin/tracking/FleetMap.tsx, components/portal/VesselTrackingMap.tsx

Workflow: Combines live vessel coordinates from terminal telemetry and manual asset registries. Renders an interactive Google Maps interface. Administrators view the entire global fleet with live status overlays. Customers see an isolated view containing only their assigned vessels with speed, heading, latitude/longitude coordinates, and last-seen timestamps.

5.9 Service Plans & Catalog Management

Implementation: lib/actions/plan-actions.ts, models/ServicePlan.ts

Workflow: Administrators configure recurring ISP and satellite subscription packages. Configurable parameters include Plan Name, Monthly Base Fee, Downlink Speed (Mbps), Uplink Speed (Mbps), Included Data Allowance (GB), Contention Ratio, and Excess Usage Rates. Subscriptions link customers to these plans with custom activation dates.

5.10 Customer Helpdesk & Support Ticketing

Implementation: lib/actions/ticket-actions.ts, components/portal/SupportTicketModal.tsx

Workflow: Provides two-way communication between customers and technical support. Customers submit tickets with Category (Billing, Technical, Account, Hardware) and Priority (Low, Medium, High, Urgent). Support agents respond in a chronological message thread, update ticket statuses (OPEN, IN_PROGRESS, RESOLVED, CLOSED), and assign staff members.

5.11 Team Management, Audit Logs & System Settings

Implementation: lib/actions/settings-actions.ts, models/ActivityLog.ts, models/Settings.ts

Workflow: Super Admins manage internal staff accounts, configure company profile parameters (Business Name, Tax Number, Address, Currency, Invoice Notes), and configure SMTP credentials for automated email dispatch. The immutable ActivityLog records every administrative operation with timestamp, actor identity, action type, and detailed payload.

6. MAIN USER WORKFLOWS

6.1 Authentication & First-Login Security Gate Lifecycle

Step 1: Admin Invitation	Admin registers customer in portal. System generates temporary password, sets mustChangePassword=true, and sends email.
Step 2: Initial Login Attempt	User navigates to /login and enters temporary credentials. System validates bcrypt hash and identifies temporary flag.
Step 3: Forced Password Gate	DAL middleware intercepts request and redirects directly to /first-login-change-password. All dashboard routes are locked.
Step 4: Password Rotation	User provides temporary password and defines a new secure password (min 8 chars, 1 digit). Hash updated, flag cleared to false.
Step 5: Session Establishment	System creates encrypted JWT session cookie (hn_session) and dispatches user to authorized dashboard (/admin or /portal).

6.2 CDR Ingestion, Rating & Unallocated Exception Lifecycle

Step 1: Wholesale Ingestion	Admin drags and drops bulk wholesale CDR CSV in Billing -> CDR Import. PapaParse parses rows and SHA-256 prevents duplicates.
Step 2: Header Auto-Detection	Dynamic header matcher maps CSV column aliases for Customer Codes, Product Identifiers, and Wholesale Amounts.
Step 3: Customer Code Resolution	Engine checks row code against all User.customerCodes arrays. If unmatched -> quarantined as UNMATCHED (Unknown Account).
Step 4: Product Rating Execution	Engine matches telecom code (e.g. CALL-123) against active Retail Plans, applies markup percentage, and snapshots in CdrChargeRecord.
Step 5: Unallocated Resolution	If unmatched rows exist, system displays Unallocated Alert Card. Admin exports CSV, adds missing customer/mapping, and clicks 'Reprocess'.

7. ADMIN WORKFLOW (STEP-BY-STEP)

A typical day-to-day administrative lifecycle follows six structured steps:

Step 1: Dashboard Inspection: Admin logs in at /login and lands on /admin. Inspects real-time statistics: Total Active Customers, Monthly Billed Revenue, Online vs Offline Terminals, and Open Support Tickets.

Step 2: Customer Onboarding & Multi-Account Setup: Admin navigates to Customers and clicks 'Add Customer'. Enters contact info, assigns primary and secondary account codes (e.g. ZZSP100, ACC-2001), links a Starlink Vessel ID, and selects a recurring Service Plan. System sends an automated invite email with temporary credentials.

Step 3: CDR Ingestion & Processing: Admin navigates to CDR Import, drags and drops the wholesale CSV file. System parses records, calculates charges, and alerts the admin if any records have missing accounts or unknown product codes.

Step 4: Exception Resolution: If unallocated rows are detected, admin downloads the Unallocated CSV Report, verifies the unknown code, creates the customer or adds the product mapping, and clicks 'Reprocess Unmatched'.

Step 5: Monthly Invoicing: Admin navigates to Invoices, clicks 'Generate Monthly Invoices'. System bundles rated CDR charges + recurring base fees, generates PDF invoices, and sends email notifications to clients.

Step 6: Fleet & Telemetry Monitoring: Admin checks the Tracking map to inspect live vessel positions and monitors terminal telemetry (latency, ping drops, SNR) across all customer vessels.

8. CUSTOMER / USER WORKFLOW (STEP-BY-STEP)

The end client self-service journey is streamlined and frictionless:

Step 1: Account Activation & First-Login Reset: Client receives invitation email with temporary password. Upon logging in, system automatically redirects to /first-login-change-password. Client sets their secure password and enters the portal.

Step 2: Unified Multi-Account Overview: Client lands on /portal/overview. The dashboard aggregates total data consumption, voice minutes, and SMS counts across all linked account numbers.

Step 3: Terminal & Starlink Health Monitoring: Client checks terminal status cards. If Starlink is enabled, live telemetry displays current latency, download/upload speeds, SNR, and obstruction alerts.

Step 4: Live Fleet GPS Map: Client navigates to /portal/tracking to view interactive Google Maps tracking displaying the real-time position, speed, and heading of their vessels.

Step 5: Invoice Review & PDF Download: Client visits /portal/invoices to inspect current and past invoices, review line-item CDR usage charges, and download official PDF tax invoices.

Step 6: Helpdesk Ticketing: If assistance is required, client clicks 'New Ticket' in /portal/support, describes the inquiry, and receives threaded replies from the support team.

9. SYSTEM ARCHITECTURE

Layer 1: Client Interfaces	Dual web portals for Administrators (/admin/*) and Corporate Clients (/portal/*) with responsive desktop and tablet layouts.
Layer 2: Edge Proxy & DAL	Next.js Middleware edge proxy with domain dispatch, encrypted JWT session parsing, role enforcement, and password change interception.
Layer 3: Application Server	Next.js 16 App Router featuring Server Components, React Server Actions, PapaParse streaming engine, and @react-pdf/renderer pipeline.
Layer 4: Database Layer	MongoDB cluster with Mongoose ORM managing 11 relational collections (Users, Subscriptions, CDR Batches, Charges, Invoices, Tickets, Logs).
Layer 5: External Services	SLASH API (Starlink live telemetry), Google Maps JavaScript SDK (fleet visualization), and Nodemailer SMTP (email dispatch).

10. DATA FLOW ARCHITECTURE

1. CDR Wholesale-to-Retail Stream:

Raw CSV File -> PapaParse Streaming Parser -> Dynamic Header Detection -> Customer Code Matcher (User.customerCodes) -> Product Identifier Matcher (RetailPlan) -> Immutable Rating Snapshot (CdrChargeRecord) -> Batch Header Summary -> Monthly Invoice Aggregation.

2. Live Starlink Telemetry Stream:

SLASH API (/v1/starlink) -> Connector with Backoff & Deduplication -> Normalization Service -> Real-time UI Gauge Widgets & Google Maps GPS Pin Overlay.

3. PDF Invoicing Stream:

Rated CDR Charges + Recurring Subscription Line Items -> Dynamic Tax Calculation -> React-PDF Layout Pipeline -> Binary Buffer Stream -> Browser Client Download.

11. DATABASE OVERVIEW & ENTITY RELATIONSHIPS

The application utilizes MongoDB via Mongoose ORM. Below is the relational mapping of core models:

Entity / Collection	Key Schema Fields	Relationships & Purpose
User	email, passwordHash, role, status, customerCode, customerCodes[] , starlinkVesselId, mustChangePassword	Central auth identity. Stores multi-account code array. Linked to Invoices, Tickets, Subscriptions.
CustomerAccount	accountNumber, accountName, contactEmail, phone, companyAddress, status	Dedicated corporate account entity mapped to customer codes.
Subscription	customerId, planId, status, staticIp, terminalId, startDate, renewalDate	Binds a User (Customer) to a recurring ServicePlan.
ServicePlan	name, monthlyPrice, downloadSpeed, uploadSpeed, dataLimitGB, contentionRatio	Master catalog of recurring ISP/Satellite subscription plans.
RetailPlan	name, planType (MARKUP / FIXED), markupPercentage, fixedPrice, status	Wholesale CDR rating configuration rules.
CdrIdentifierMapping	identifier, trafficType, retailPlanId, description, isActive	Maps raw CDR product codes (CALL-123, SMS-245) to RetailPlans.
CdrImportBatch	fileName, fileHash, totalRows, matchedRows, unmatchedRows, wholesaleTotal, retailTotal	Header record for bulk or single CDR upload operations.
CdrChargeRecord	batchId, customerId, customerCode, productCode, wholesaleAmount, calculatedRetail, status	Itemized rated CDR transaction. Linked to Invoices for billing.
Invoice	invoiceNumber, customerId, billingPeriod, subtotal, taxAmount, totalAmount, status, lineItems[]	Master billing record. Generates downloadable PDF invoices.
SupportTicket	ticketNumber, customerId, subject, category, priority, status, assignedTo, messages[]	Customer helpdesk ticketing and threaded communications.
ActivityLog	actorId, actorRole, action, targetCustomer, metadata, ipAddress, timestamp	Immutable system audit trail of all administrative actions.

12. AUTHENTICATION & SECURITY CONTROLS

- **Cryptographic Password Hashing:** Passwords are never stored in plaintext. They are salted and hashed using bcryptjs (10 rounds).
- **Encrypted JWT Session Cookies:** Sessions use signed, encrypted JWT tokens transmitted via HTTP-only, SameSite=Lax cookies (hn_session).
- **Brute-Force Account Protection:** Enforces rate limiting and account locking after 5 consecutive failed login attempts (15-minute lockout).
- **Mandatory First-Login Password Gate:** Temporary passwords generated by admins are restricted by mustChangePassword=true, preventing all portal access until rotated.
- **Edge Middleware & DAL Authorization:** Every server action and page route validates user roles at the Data Access Layer before executing database operations.
- **Comprehensive Audit Logging:** Sensitive actions (user creation, password updates, invoice generation, CDR uploads) write immutable records to ActivityLog.

13. EXTERNAL INTEGRATIONS

External Service	Integration Architecture	Operational Functionality
SLASH API (Starlink Telemetry)	REST API (https://slash-api.rudra.sh/api/v1) Authenticated via X-API-Key	Pulls real-time Starlink terminal latency, ping drops, bandwidth, and obstruction metrics.
Google Maps JavaScript API	Client-side Maps SDK Custom SVG vessel pins & heading vectors	Renders live GPS coordinates, vessel headings, speed, and fleet positions globally.
SMTP Email Relay (Nodemailer)	TLS SMTP Transport Configured in System Settings	Dispatches user invitations, temporary credentials, password reset links, and invoice notices.
PapaParse & React-PDF	In-memory streaming & buffer generation Zero external cloud dependency	Parses multi-megabyte CDR files in browser/server and generates branded PDF invoices.

14. REPORTS & DATA MANAGEMENT

- **Unallocated CDR Exception Report (.csv):** Generated on-demand via /api/admin/cdr-import/[batchId]/unallocated-report. Provides row-by-row analysis of unmatched records with original row index, missing Customer Code, unmapped Product Code, wholesale amount, and rejection reason.
- **Batch CDR Reconciliation Report (.csv):** Full export of all processed records in an import batch, showing both wholesale inputs and calculated retail charges.
- **Tax-Compliant PDF Invoices:** Formatted with corporate letterhead, billing periods, payment remittance details, and itemized data/voice/SMS usage.
- **Fleet Inventory & Usage Audits:** Comprehensive tabular views and exportable datasets of active satellite terminals, IP allocations, and customer usage.

15. IMPLEMENTATION STATUS MATRIX

Direct feature-by-feature verification against actual codebase implementation:

Feature / Capability	Implementation Status	Codebase Location / Module
Dual-Portal Architecture	● Production Ready	proxy.ts, app/admin, app/portal
Authentication & Session DAL	● Production Ready	lib/auth/dal.ts, lib/auth/session.ts
First-Login Forced Password Reset	● Production Ready	app/first-login-change-password
Multi-Account Customer Linking	● Production Ready	models/User.ts (customerCodes[])
Wholesale CDR CSV Ingestion	● Production Ready	lib/cdr/parser.ts, CdrUploadZone.tsx
Dynamic Column Header Detection	● Production Ready	lib/cdr/parser.ts (detectColumns())
Compound Product Code Rating	● Production Ready	lib/billing/pricing-engine.ts
Unallocated CDR Alert & CSV Export	● Production Ready	api/admin/cdr-import/.../unallocated-report
One-Click CDR Reprocessing	● Production Ready	lib/actions/cdr-actions.ts
Automated Invoice Generation	● Production Ready	lib/actions/invoice-actions.ts
Dynamic PDF Invoice Rendering	● Production Ready	lib/pdf/invoice-template.tsx
Starlink SLASH Telemetry Client	● Production Ready	lib/starlink/client.ts
Interactive GPS Fleet Tracking	● Production Ready	components/admin/tracking/FleetMap.tsx
Support Helpdesk & Threading	● Production Ready	lib/actions/ticket-actions.ts
Immutable Activity Audit Logging	● Production Ready	models/ActivityLog.ts

16. CURRENT LIMITATIONS

- **Manual Payment Status Tracking:** Invoices currently support status transitions (Draft, Sent, Paid, Void) marked by administrators (e.g. Bank Wire / Direct Transfer) rather than an embedded instant credit card payment gateway.
- **External Telemetry Upstream Dependency:** Live Starlink metrics require active vessel linkage and valid credentials for the upstream SLASH API service.
- **Manual CSV Upload Trigger:** Wholesale CDR files are uploaded through the web UI rather than an automated SFTP server polling daemon.

17. FUTURE IMPROVEMENT ROADMAP

- **Integrated Payment Gateway:** Embedded Stripe / Adyen checkout for instant customer invoice settlement and recurring auto-pay credit card billing.
- **Automated SFTP / S3 CDR Ingestion:** Background cron scheduler to automatically pull and rate CDR files from satellite wholesale FTP dropboxes.
- **SMS & Webhook Alerting:** Instant SMS notifications (via Twilio/AWS SNS) when vessels exceed bandwidth allowances or experience satellite link drops.
- **Advanced Telemetry AI Diagnostics:** Machine learning anomaly detection to predict terminal hardware failures or antenna misalignment based on historical SNR patterns.

18. FINAL SUMMARY & VERIFICATION

PLATFORM VERIFICATION & PRODUCTION READINESS

The **Hybrid Networks Portal** is a fully verified, production-grade enterprise application. Every feature documented in this manual—including multi-account customer linking, fuzzy CDR column detection, wholesale-to-retail markup rating, unallocated exception reporting, React-PDF invoice generation, real-time SLASH Starlink telemetry, interactive Google Maps fleet tracking, and mandatory first-login password security—is fully implemented and operational in the codebase.

This platform provides a robust foundation for satellite service providers, maritime fleet operators, and enterprise telecommunications management.

Prepared By:	Enterprise QA & Engineering Team	Client:	Hybrid Networks & Enterprise Clients
Status:	● Production Ready	Date:	September 2026